CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Paper Bridge

Goal: Transform paper into a bridge that spans a gap and carries increasing loads.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 6

Ask | Plan | Build | Test | Improve

You are done when

Report the maximum load and explain how one shape helped carry it.

You will use

- Copy paper
- Index cards
- Tape
- Two support blocks
- Uniform load pieces
- Recording sheet

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place the support blocks on the marks.
- Count the load pieces.

2 Step 2: Test safely — 3 min

- Keep hands below the table edge.
- Place only approved loads on the bridge.

3 Step 3: Plan the span — 5 min

- Choose a beam shape.
- Predict where the paper may bend.
- Sketch the bridge.

4 Step 4: Build the bridge — 11 min

- Fold the paper into your chosen shape.
- Tape only the planned joints.
- Place the bridge across the fixed span.

5 Step 5: Load the bridge — 7 min

- Add one load piece.
- Record the supported total.
- Repeat until the bridge fails.
- Change one weak part.

6 Step 6: Reset the test — 3 min

- Return every load piece.
- Recycle damaged paper.

7 Step 7: Exit check — 3 min

- Write the highest supported load.
- Name your useful change.

What to turn in

Your record sheet: most load held, one rebuild, same test repeated.

No computer? Another way to do it

The teacher runs one class test while teams predict and record.

Finished early? Next challenge

Efficiency Engineer: Remove one piece of material while keeping the bridge able to hold its target load.

Stay safe

Use only the provided loads and keep hands and feet clear beneath the test span.

Grade targets

Grade 5: Record the most load held with tally marks. Rebuild one part.
Grade 6: Record loads in a table. Explain which shape change mattered and why.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.